MACHINE LEARNING & GENERATIVE AI — MODULE 6

Key Challenges & Limitations of Generative AI

University-Exam & Interview-Ready Study Notes

Topics Covered:

1. Prompt Sensitivity
2. Structured vs Unstructured Data
3. Knowledge Cutoff
4. Non-Deterministic Nature of LLMs
5. Hallucinations
6. Lack of Common Sense
7. Bias and Fairness
8. Data Privacy, Security & Misuse
9. Explainable AI, Responsible AI, Ethics & Governance
10. Real-Life Case Studies

Prepared for Coursera Machine Learning & Generative AI Course — Module 6 Notes

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TOPIC 1

Prompt Sensitivity

1. Definition

Prompt Sensitivity means an AI model can give very **different answers** depending on small changes in how a question (prompt) is worded — even if the meaning seems the same to a human.

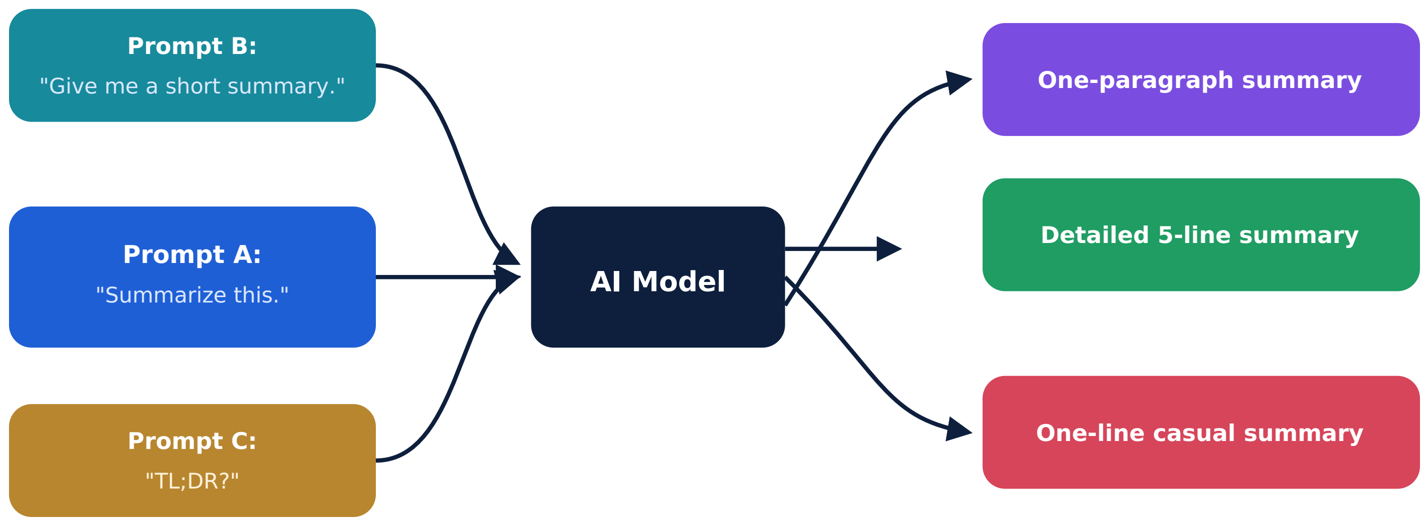
2. Why Does Prompt Wording Matter?

An LLM doesn't "understand" a question the way a human does — it predicts the next words based on patterns in its training data. Small wording changes shift which patterns the model matches, so the phrasing, word order, and even punctuation can change the response.

3–4. How It Happens & Step-by-Step

1. You type a prompt using specific words and structure.
2. The model converts your prompt into tokens (see Module 5 notes).
3. It predicts a response based on patterns most associated with that exact wording.
4. A differently-worded prompt — even with the same intended meaning — activates slightly different patterns, producing a different response.

Prompt → AI → Different Outputs



Three prompts with similar intent can still produce noticeably different outputs.

Good, Bad & Ambiguous Prompts

Type	Example	Result
Good Prompt	"Summarize this article in 3 bullet points for a beginner."	Clear, specific, consistent output
Bad Prompt	"Do the thing with this."	Confusing, unpredictable output
Ambiguous Prompt	"Tell me about Python."	Could mean the programming language OR the snake

Prompt Engineering & Prompt Optimization

Prompt Engineering is the practice of carefully designing prompts to reliably get better, more predictable results. **Prompt Optimization** is the process of testing and refining prompt wording over time to consistently improve output quality.

5. Real-Life Examples

ChatGPT/Claude — Asking "explain like I'm 5" vs "explain in technical detail" gives very different depth of explanation.

Customer Support Chatbots — A vaguely worded customer question can lead the bot to a wrong department or irrelevant help.

Education — Students get very different quality answers depending on how precisely they phrase a study question.

E-commerce Search Assistants — "cheap laptop" vs "budget laptop under $500 for students" retrieves very different product results.

6–7. Advantages & Limitations

Advantages of Understanding Prompt Sensitivity	Limitations It Causes
Lets skilled users fine-tune outputs precisely with better prompts	Inconsistent results for casual/inexperienced users
Encourages clearer human communication	Makes AI behaviour harder to fully predict

8. How Researchers Solve It

- Training models to be more robust to small wording differences.

- Building instruction-tuned models that better generalise across phrasings.
- Studying and publishing prompting guides based on empirical testing.

9. Best Practices (For Developers & Users)

- Be specific about format, length, and audience in your prompt.
- Test prompts with multiple phrasings before relying on them in production.
- Use system prompts to set consistent behaviour across a whole application.

10. Interview Questions

Q1. Why does an LLM respond differently to two prompts that mean the same thing?
A: The model predicts text based on patterns learned from training data; different wordings activate different learned patterns, even if human intent is the same.

Q2. What is Prompt Engineering, and why is it useful?
A: It's the practice of carefully crafting prompts to get more reliable, higher-quality outputs — useful because it reduces the unpredictability caused by prompt sensitivity.

Q3. How can developers reduce the impact of prompt sensitivity in a product?
A: By using well-tested system prompts, providing prompt templates, and testing across many phrasings before deployment.

11. Practice Questions

- Rewrite an ambiguous prompt into a clear, specific one.
- Explain why "Tell me about Python" is an ambiguous prompt.
- What's the difference between prompt engineering and prompt optimization?

Memory Trick

```
"Same meaning, different words → different AI answer."
One-liner: Small wording change = possibly big output change.
```

12. Summary

Prompt Sensitivity means an AI's output can shift significantly based on exact wording, since the model matches learned patterns rather than truly understanding meaning — careful prompt engineering helps manage this.

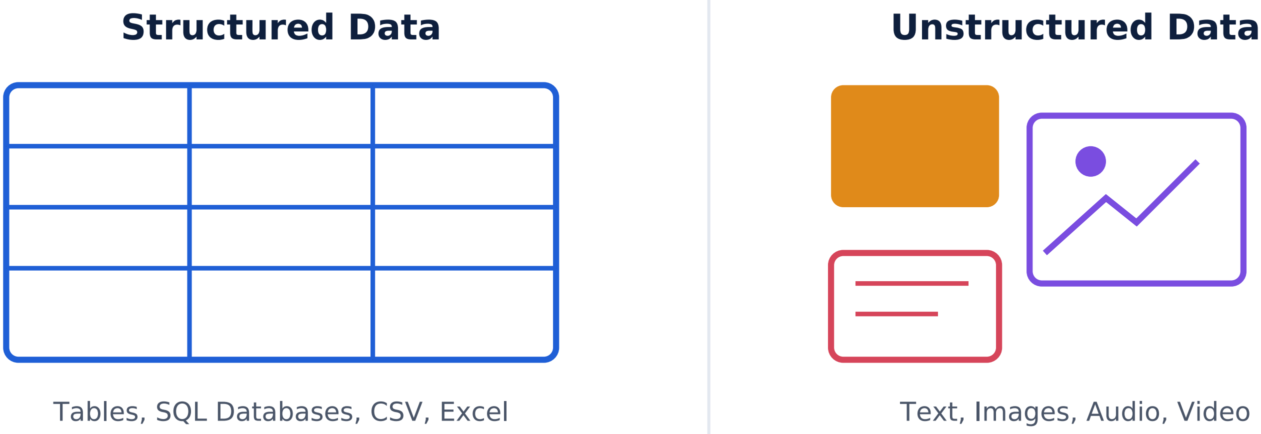
TOPIC 2

Structured vs Unstructured Data

1. Definition

> **Structured Data** is information organised in a fixed format, like rows and columns (tables), that computers can easily search and analyse. **Unstructured Data** has no fixed format — like free text, images, or audio.

Structured vs Unstructured — Visual Comparison



Structured data fits neatly into tables; unstructured data comes in free-form types.

Comparison Table

Aspect	Structured Data	Unstructured Data
Format	Fixed rows and columns	Free-form, no fixed schema
Examples	SQL databases, CSV files, Excel spreadsheets	Emails, images, audio, social media posts
Ease of Analysis	Easy with traditional tools (SQL queries)	Needs AI/ML (especially Deep Learning) to interpret
Storage	Relational databases	Data lakes, cloud object storage
Percentage of World's Data	Roughly 20%	Roughly 80%

Why Structured Data Matters for AI

- Easier and faster to feed into traditional Machine Learning models.
- Enables precise filtering, sorting, and querying (e.g., "show all customers over 30").
- Often used alongside unstructured data in real systems (e.g., a database of customer records plus their free-text support tickets).

5. Real-Life Examples

Banking — Structured: account balances table. Unstructured: customer complaint emails.

Healthcare — Structured: lab results database. Unstructured: doctor's handwritten notes.

E-commerce — Structured: product price/inventory tables. Unstructured: customer reviews.

6–7. Advantages & Limitations

Structured Data	Unstructured Data
+ Easy to search/query, - Limited to predefined fields	+ Captures rich real-world detail, - Harder and more expensive to analyse

Interview Questions

Q1. Why is most of the world's data unstructured?

A: Because real-world communication (text, images, audio, video) doesn't naturally fit into rows and columns — it's naturally free-form.

Q2. Why do modern AI systems often need both types of data?

A: Structured data provides clean, precise facts (like account numbers), while unstructured data captures nuance and context (like customer sentiment) — combining both gives a fuller picture.

Memory Trick

```
"Structured = Spreadsheet. Unstructured = Everything else."
```

Summary

> Structured data fits neatly into tables/databases and is easy to query; unstructured data (text, images, audio) makes up most of the world's data and requires AI to interpret effectively.

TOPIC 3

Knowledge Cutoff

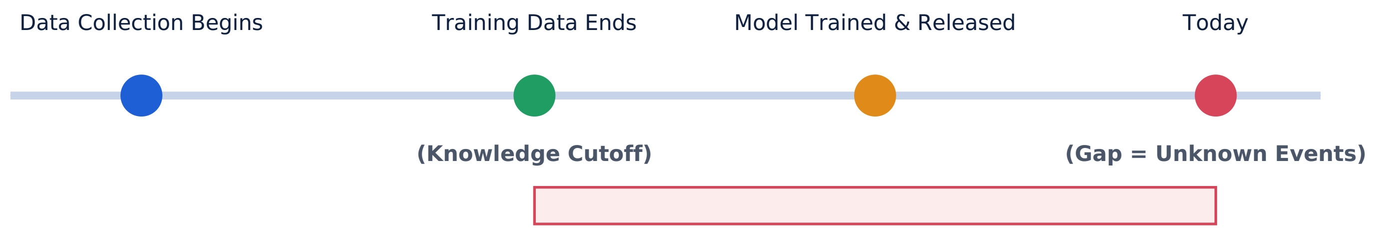
1. Definition

Knowledge Cutoff is the date after which an AI model has no built-in knowledge — because its training data was collected up to that point only.

2–3. Why AI Has Outdated Knowledge & Why It Happens

Training a large AI model takes months and enormous computing resources. The data used is collected up to a certain date, then the model is trained on it. Once training finishes, the model doesn't automatically learn about anything that happened afterward — its "knowledge" is frozen at that snapshot in time.

Timeline Diagram



Everything after the Knowledge Cutoff is a "blind spot" for the model unless given external tools.

How This Gap Is Closed

Technique	How It Helps
Fine-tuning	Re-training the model on newer, additional data to update some knowledge.
Retrieval-Augmented Generation (RAG)	Lets the model retrieve current documents at answer time, bypassing the frozen cutoff.
Web Search Integration	Connects the model directly to live web search results for up-to-date answers.

5. Real-World Examples

ChatGPT/Claude/Gemini — Without search enabled, may not know about very recent news or events.

Banking — A model without live data access can't know today's exact interest rates unless connected to a live system.

Customer Support — An AI assistant needs RAG or live database access to know about a product launched last week.

Limitations

- Cannot answer questions about events after its training cutoff without external tools.
- May confidently give outdated information if not corrected by retrieval or search.

Interview Questions

Q1. Why can't an LLM simply "learn" about today's news automatically?

A: Its knowledge comes only from its fixed training dataset; without a retrieval or search mechanism, it has no way to access anything published after training ended.

Q2. What are two ways to overcome knowledge cutoff limitations?

A: Retrieval-Augmented Generation (RAG) and live web search integration — both let the model access current information without retraining.

Memory Trick

```
"Cutoff = Model's memory freezes on a specific date."
```

Summary

Knowledge Cutoff means an LLM's knowledge is frozen at its training data's collection date; RAG, web search, and fine-tuning help bridge this gap with current information.

TOPIC 4

Non-Deterministic Nature of LLMs

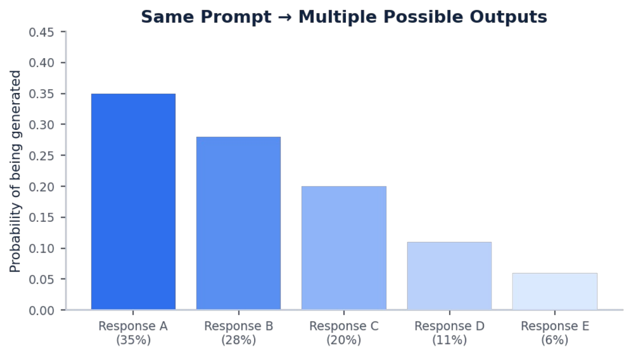
1. Definition

Deterministic means the same input always produces exactly the same output. **Non-deterministic** means the same input can produce different outputs each time — which is how most LLMs behave by default.

2–3. Why AI Gives Different Answers

LLMs generate text by predicting a probability for many possible next tokens, then **sampling** (randomly picking, weighted by probability) rather than always choosing the single most likely one. This controlled randomness is what allows varied, creative-sounding responses — but it also means asking the same question twice can give different answers.

Probability Diagram



The same prompt can lead to different responses, each with its own probability of being generated.

Randomness, Temperature & Sampling

Term	Meaning
Randomness	The element of chance introduced when selecting the next token.
Temperature	A setting controlling how much randomness is used — low temperature = more consistent, high = more varied.
Sampling	The method of randomly choosing a token weighted by its probability, instead of always picking the top one.

Creativity vs Consistency

Higher randomness (temperature) increases creativity and variety but reduces consistency; lower randomness increases reliability and repeatability but can feel more repetitive.

5. Real-Life Examples

ChatGPT/Claude — Asking the same creative writing prompt twice can produce two different stories.

Customer Support Bots — Companies often set low temperature so answers stay consistent and predictable.

Education — A student asking the same question twice might get slightly different (but similarly correct) explanations.

Advantages & Limitations

Advantages	Limitations
Enables creative, varied writing and brainstorming	Makes outputs harder to test/reproduce exactly
Avoids overly repetitive, robotic answers	Can reduce trust in critical/factual applications

Best Practices

- Use low temperature (or temperature = 0) for tasks needing consistency, like factual Q&A or code generation.
- Use higher temperature for creative writing or brainstorming tasks.

Interview Questions

Q1. Why are LLMs typically non-deterministic?

A: They sample the next token from a probability distribution rather than always choosing the single highest-probability token, introducing controlled randomness.

Q2. How would you make an LLM's output more consistent?

A: Lower the temperature setting (or use greedy/deterministic decoding), which reduces randomness in token selection.

Memory Trick

`"Same question, different day, maybe different answer — that's non-determinism."`

Summary

LLMs are non-deterministic because they sample from probability distributions rather than always picking the same top choice; temperature controls the balance between creativity and consistency.

TOPIC 5

Hallucinations

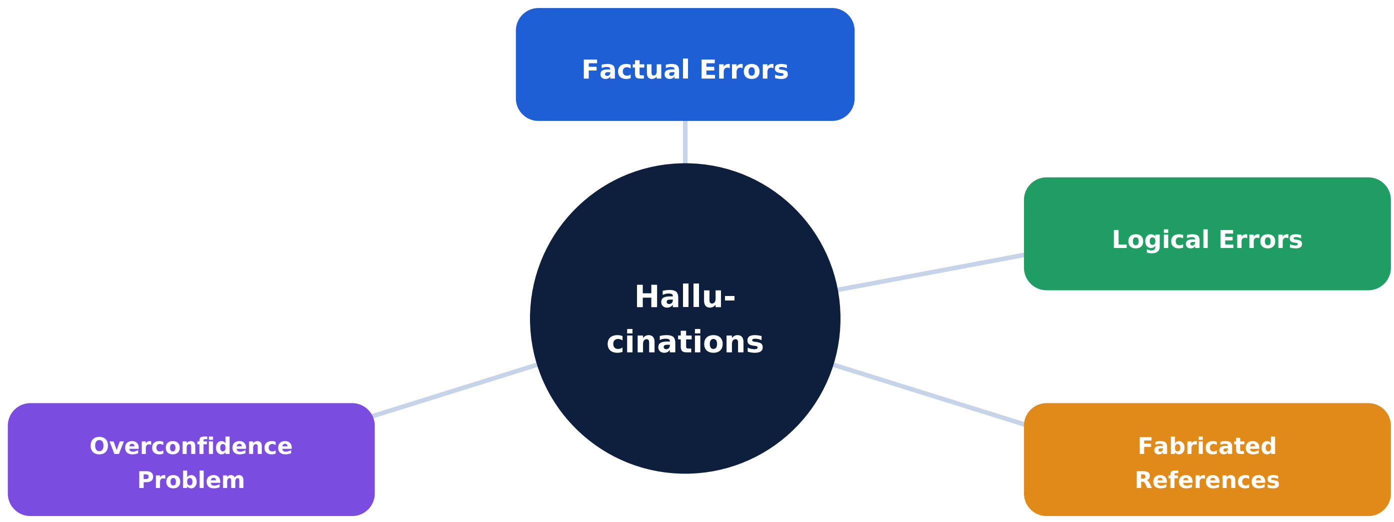
1. Definition

A hallucination is when an AI model confidently generates information that is false, made-up, or not actually supported by real facts — while sounding completely plausible.

2–3. Why Hallucinations Occur & How They Happen

An LLM generates text by predicting the statistically most likely next words based on patterns — it has no built-in fact-checking system. If a query touches an area with sparse, conflicting, or missing training data, the model may still "fill in the gap" with a fluent-sounding but incorrect answer, because generating plausible text is its core skill, not verifying truth.

Types of Hallucination — Mind-Map



Common categories of hallucination in AI-generated text.

Types Explained

Type	Meaning	Example
Factual Hallucination	Stating an incorrect fact confidently	Giving the wrong birth year for a historical figure
Logical Hallucination	Reasoning that doesn't logically follow	A math "proof" with a step that doesn't actually follow
Fabricated References	Inventing fake sources, citations, or quotes	Citing a research paper that doesn't exist
AI Confidence Problem	Sounding equally confident whether right or wrong	Stating a made-up statistic as if it were verified fact

5. Real-Life Examples

ChatGPT — Has been documented inventing plausible-sounding but fake legal case citations.

Google Gemini — Has produced factually incorrect summaries of search results in some cases.

Healthcare — An AI assistant confidently suggesting an incorrect drug dosage if not properly grounded.

Education — Students receiving fabricated book/article references when asking for sources.

Limitations Caused by Hallucination

- Reduces trust in AI-generated content, especially for factual or high-stakes tasks.
- Can spread misinformation if outputs aren't fact-checked by a human.

8. How Researchers Reduce Hallucinations — Architecture

Combining RAG and fact-checking layers reduces (but doesn't eliminate) hallucination risk.

- **RAG** — grounding answers in retrieved, real documents.
- **RLHF** — training the model to be more cautious and honest using human feedback.
- **Confidence calibration** — teaching models to better express uncertainty instead of guessing confidently.

9. Best Practices for Developers

- Always fact-check AI outputs for high-stakes use cases (medical, legal, financial).
- Use RAG to ground responses in verified data sources.
- Encourage users to verify citations and sources independently.

Interview Questions

Q1. Why do LLMs hallucinate instead of simply saying "I don't know"?
A: They are trained to generate the most statistically likely continuation of text, not to verify truth — without explicit training or tools to recognise uncertainty, they may generate a plausible-sounding guess instead.

Q2. How does RAG help reduce hallucination?
A: By supplying the model with real, retrieved documents as context, grounding its answer in verifiable facts rather than relying purely on memorised patterns.

Q3. Give an example of a fabricated reference hallucination.
A: An AI model citing a specific academic paper, with a title and authors, that does not actually exist.

Memory Trick

```
"Hallucination = Confident + Wrong."
```

Summary

Hallucinations occur because LLMs generate statistically plausible text rather than verified facts; RAG, fact-checking, and RLHF help reduce (but not fully eliminate) this risk.

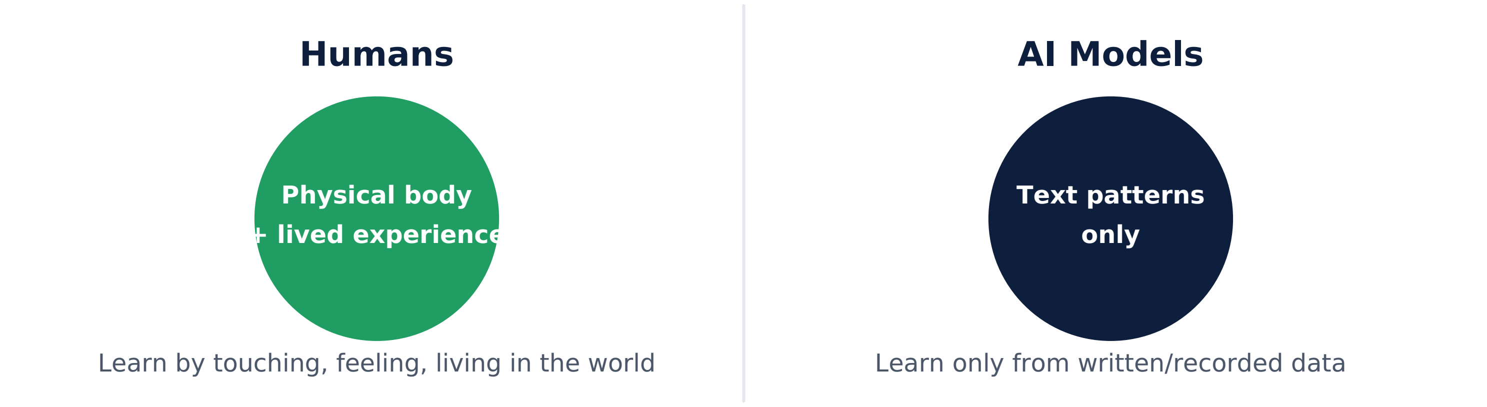
TOPIC 6

Lack of Common Sense

1. Definition

Common sense is the practical, everyday understanding of how the physical and social world works — the kind of obvious knowledge humans pick up naturally through lived experience, which AI models often lack.

Why Humans Have Common Sense (and AI Doesn't)



Humans learn common sense by physically experiencing the world; AI only learns from text/data patterns.

2–3. Physical Reasoning & Everyday Reasoning

- **Physical Reasoning** — understanding physical cause and effect, like "if you drop a glass, it might break."
- **Everyday Reasoning** — understanding normal social/practical situations, like "you shouldn't wear a swimsuit to a job interview."

AI models learn from text describing the world, not from directly experiencing it — so they can miss obvious physical or social facts that were never explicitly written down, because humans rarely bother to state the truly obvious.

5. Practical & Real-Life Examples

Self-Driving Cars — May struggle with unusual real-world situations a human driver would instinctively handle (e.g., an oddly-dressed pedestrian).

Chatbots — Might not realise a request like "put the fridge in the microwave" is physically nonsensical.

Healthcare AI — May miss obvious practical context a nurse would instantly notice in a real patient interaction.

Customer Support — A bot might not realise a sarcastic complaint is not a literal request.

Limitations

- Can produce technically-worded but practically absurd suggestions.
- Struggles with situations requiring physical intuition or lived social context.

How Researchers Address This

- Building specialised "common sense" reasoning datasets and benchmarks.
- Combining language models with robotics/physical simulation for embodied learning.
- Using human feedback to correct nonsensical outputs.

Interview Questions

Q1. Why do AI models often lack common sense despite vast training data?
A: They learn only from text/data patterns, not from direct physical or social experience, so implicit "obvious" knowledge that's rarely written down explicitly is often missing from their training.

Q2. Give an example of a common-sense failure in a self-driving car context.
A: Misjudging an unusual but harmless situation (like a person in a costume) as a serious hazard, or vice versa, due to lack of true real-world intuition.

Memory Trick

`"AI reads about the world; humans live in it."`

Summary

AI lacks common sense because it learns from text/data patterns rather than lived physical and social experience, which can lead to practically nonsensical outputs in edge cases.

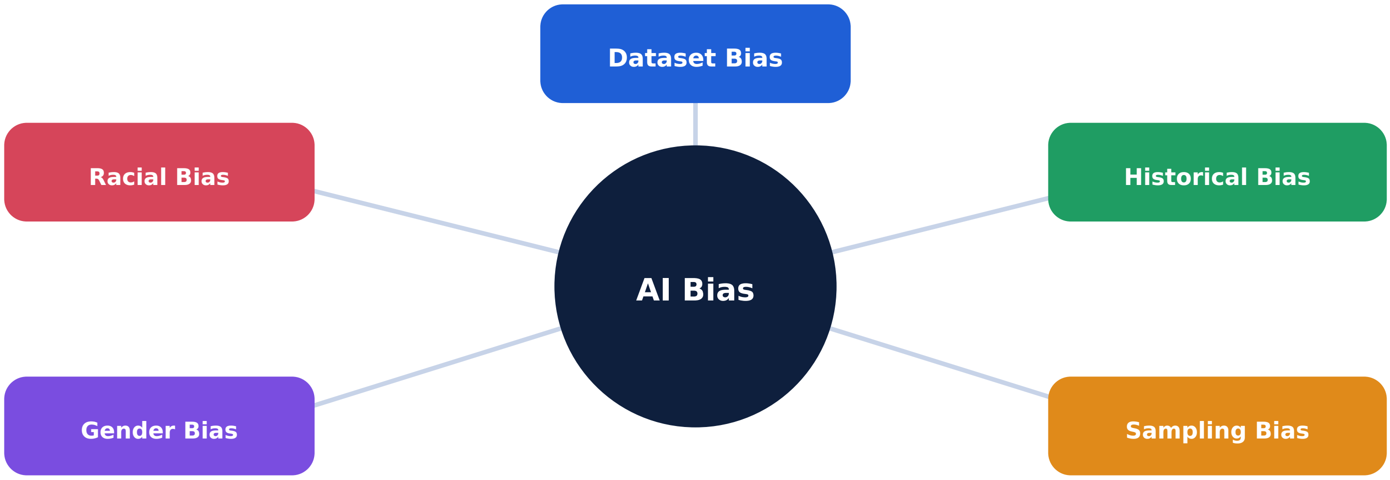
TOPIC 7

Bias and Fairness

1. Definition

AI Bias means a model's outputs unfairly favour or disfavour certain groups of people, usually because of patterns picked up from imbalanced or unfair training data. **Fairness** is the goal of making AI decisions equitable across different groups.

Why Bias Occurs — Types



Different sources of bias that can creep into an AI system.

Bias Types Explained

Type	Meaning
Dataset Bias	Training data doesn't represent all groups fairly.
Historical Bias	Past human decisions (already unfair) get baked into the data the model learns from.
Sampling Bias	Data collection methods overrepresent or underrepresent certain groups.
Gender Bias	Model favours or stereotypes one gender over another.
Racial Bias	Model treats different racial groups unfairly or inaccurately.

Biased vs Fair Outcome — Comparison

Aspect	Biased System	Fair System
Training Data	Skewed toward one group	Balanced, representative of all relevant groups
Outcome	Systematically favours/disfavours a group	Equal treatment/opportunity across groups
Example	Hiring AI favouring one gender due to biased historical data	Hiring AI evaluated and adjusted for equal outcomes across genders

5. Real-Life Examples

AI Hiring Tools — Some past hiring algorithms downgraded resumes with certain gendered keywords due to historically skewed training data.

Facial Recognition — Some systems have shown lower accuracy for certain skin tones due to imbalanced training datasets.

Banking — Credit scoring models can unintentionally disadvantage certain neighbourhoods if trained on historically biased lending data.

Social Media — Recommendation algorithms can reinforce stereotypes if trained on biased engagement patterns.

Mitigation Techniques (How Researchers Solve It)

- Auditing datasets for representation gaps before training.
- Using fairness-aware algorithms that actively balance outcomes across groups.
- Ongoing bias testing and monitoring after deployment.
- Diverse teams involved in AI design and review.

Best Practices

- Test model outputs across different demographic groups before release.
- Document dataset sources and known limitations transparently.
- Include human review for high-stakes decisions (hiring, lending, healthcare).

Interview Questions

Q1. What is the difference between dataset bias and historical bias?

A: Dataset bias is about the data itself lacking fair representation; historical bias is when past unfair human decisions get encoded into the data the model learns from.

Q2. How can developers reduce bias in an AI hiring tool?
A: By auditing training data for representation gaps, testing outcomes across demographic groups, and including human oversight in final hiring decisions.

Q3. What does "fairness" mean in the context of AI?
A: Ensuring AI decisions and outcomes don't systematically disadvantage any particular group, based on characteristics like gender, race, or age.

Memory Trick

```
"Biased data in → biased decisions out."
```

Summary

AI bias arises from unbalanced or historically unfair training data; achieving fairness requires careful data auditing, fairness-aware algorithms, and ongoing human oversight.

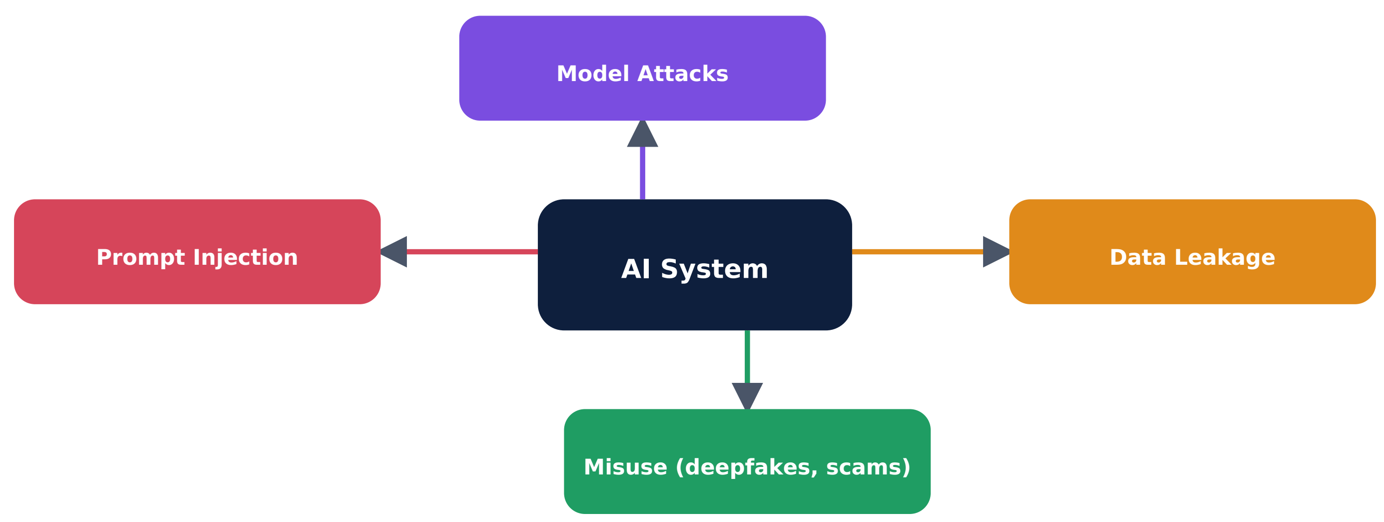
TOPIC 8

Data Privacy, Security & Misuse

1. Definitions

Data Privacy is about protecting people's personal/sensitive information from being exposed or misused. **Security** is about protecting AI systems themselves from attacks. **Misuse** refers to people using AI tools for harmful purposes.

Architecture Diagram: Risks Around an AI System



Common attack vectors and misuse risks surrounding AI systems.

Key Concepts

Term	Meaning
Personal Information	Data identifying an individual (name, address, ID numbers).
Sensitive Data	Especially private data (health records, financial info, biometric data).
Prompt Injection	A malicious input tricking an AI system into ignoring its instructions or leaking data.
Data Leakage	Private/sensitive information being unintentionally exposed by a model's output.
Model Attacks	Attempts to manipulate or extract information from a model through crafted inputs.

Types of Misuse

Misuse	Description
Deepfakes	AI-generated fake videos/audio that look/sound real.
Fake News	AI-generated misinformation spread as if factual.
Phishing/Scams	AI-crafted convincing fraudulent messages.
Academic Cheating	Using AI to complete assignments dishonestly.
Malware Generation	Attempting to use AI to help write malicious code.

5. Real-Life Examples

Banking — AI systems must avoid leaking customer account details in responses.

Social Media — Deepfake videos of public figures spreading misinformation.

Education — Schools developing policies around AI-assisted cheating.

Customer Support — Chatbots must be protected against prompt injection attempts to leak internal data.

Prevention Techniques

- Data anonymisation and encryption to protect personal information.
- Input validation and filtering to guard against prompt injection.
- Watermarking and detection tools for AI-generated content (like deepfakes).
- Usage policies, rate limits, and monitoring to catch misuse patterns.

Best Practices for Developers

- Never expose sensitive training data directly through model outputs.
- Apply strict access controls and input sanitisation.
- Regularly test systems against known attack techniques (red-teaming).

Interview Questions

Q1. What is prompt injection, and why is it a security risk?

A: It's a technique where malicious input tries to override an AI system's original instructions, potentially causing it to leak data or behave unsafely.

Q2. How can companies reduce the risk of data leakage from an AI model?

A: By carefully filtering training data for sensitive information, applying strict output controls, and testing for leakage before deployment.

Q3. What is a deepfake, and why is it concerning?

A: An AI-generated fake video or audio clip that looks/sounds real; it's concerning because it can spread convincing misinformation or be used for fraud.

Memory Trick

```
"Privacy = protect data. Security = protect systems. Misuse = bad actors abusing AI."
```

Summary

AI systems face privacy risks (data exposure), security risks (prompt injection, model attacks), and misuse risks (deepfakes, scams) — addressed through encryption, input validation, monitoring, and clear usage policies.

TOPIC 9

Explainable AI (XAI) & Responsible AI

What Is Explainable AI (XAI)?

> Explainable AI refers to techniques that help humans understand **why** an AI model made a particular decision, instead of treating it as an unexplainable "black box."

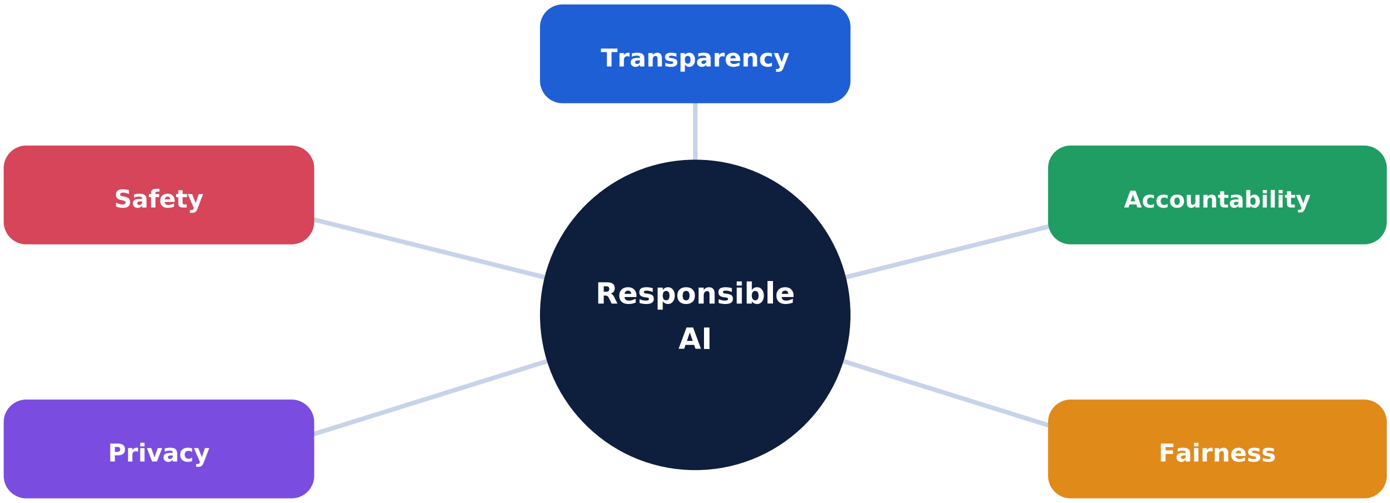
Why XAI Is Important

- Builds trust between users and AI systems.
- Required in regulated industries (finance, healthcare) to justify decisions.
- Helps developers debug and improve models.
- Helps detect hidden bias or errors in a model's reasoning.

Common XAI Methods

Method	What It Does
Feature Importance	Shows which input features most influenced a prediction.
Attention Visualization	Shows which parts of the input the model "focused on" most.
Simplified Surrogate Models	Uses a simpler, interpretable model to approximate and explain a complex one.

Responsible AI — Mind-Map



The five pillars of Responsible AI development.

Pillars Explained

Pillar	Meaning
Transparency	Being open about how an AI system works and its limitations.
Accountability	Clear responsibility for AI decisions and outcomes, held by people/organisations.
Fairness	Ensuring AI doesn't systematically disadvantage any group.
Privacy	Protecting personal data used by or generated through AI.
Safety	Preventing AI from causing harm, physically or socially.

Real-World Examples

Banking — Loan denial systems must be explainable to comply with financial regulations.

Healthcare — Diagnostic AI tools need clear explanations doctors can verify before trusting a suggestion.

Interview Questions

Q1. Why is Explainable AI especially important in healthcare and banking?

A: These fields require justifiable, auditable decisions due to regulations and the high stakes involved — doctors and regulators need to understand why an AI reached a conclusion.

Q2. Name the five pillars of Responsible AI.

A: Transparency, Accountability, Fairness, Privacy, and Safety.

Memory Trick

```
"TAFPS" = Transparency, Accountability, Fairness, Privacy, Safety — pillars of Responsible AI.
```

Summary

Explainable AI makes model decisions understandable to humans, while Responsible AI ensures systems are built and used with transparency, accountability, fairness, privacy, and safety in mind.

TOPIC 10

AI Ethics & AI Governance

Ethical AI

Ethical AI means designing and using AI systems in ways that respect human rights, avoid harm, and align with shared moral values — not just what is technically possible.

Key Concepts

Concept	Meaning
Human Oversight	Keeping humans in the loop for important AI-driven decisions, rather than full automation.
AI Regulations	Laws and rules governments create to control how AI can be developed and used.
Responsible Deployment	Carefully testing and monitoring AI systems before and after releasing them to the public.

AI Governance

AI Governance is the overall system of policies, oversight, and processes organisations use to ensure AI is developed and used safely, ethically, and legally.

AI Governance — Key Components

Component	Meaning
Policies	Internal/external rules defining acceptable AI use.
Compliance	Making sure AI systems follow relevant laws and industry standards.
Monitoring	Continuously checking deployed AI systems for problems (bias, errors, misuse).
Risk Management	Identifying and reducing potential harms before they occur.

Real-World Examples

Government Regulation — The EU AI Act sets rules classifying AI systems by risk level and requiring safeguards for high-risk uses.

Corporate Governance — Companies establishing internal AI ethics boards to review new AI products before launch.

Healthcare — Hospitals requiring human doctor sign-off on any AI-assisted diagnosis before action is taken.

Interview Questions

Q1. What is the difference between AI Ethics and AI Governance?

A: AI Ethics is about the moral principles guiding how AI should behave; AI Governance is the practical system of policies, compliance, and monitoring used to enforce those principles in practice.

Q2. Why is human oversight considered important in AI systems?

A: It ensures a person remains accountable for important decisions and can catch errors or unethical outcomes that a fully automated system might miss.

Memory Trick

```
"Ethics = the moral compass. Governance = the rules and checks that enforce it."
```

Summary

AI Ethics defines the moral principles for responsible AI behaviour, while AI Governance provides the policies, compliance checks, and monitoring needed to enforce those principles in real organisations.

REFERENCE

Comparison Tables — Quick Reference

Structured vs Unstructured Data

Aspect	Structured	Unstructured
Format	Rows/columns	Free-form
Example	SQL database	Emails, images, audio

Hallucination vs Knowledge Cutoff

Aspect	Hallucination	Knowledge Cutoff
Cause	Model generates plausible but false content	Model simply has no data past a certain date
Fix	RAG, fact-checking, RLHF	RAG, web search, fine-tuning

Deterministic vs Non-Deterministic

Aspect	Deterministic	Non-Deterministic
Output	Always the same for the same input	Can vary for the same input
Example	Basic calculator	Default LLM chat response

Human Intelligence vs AI Reasoning

Aspect	Human Intelligence	AI Reasoning
Learning Source	Lived, physical, social experience	Text/data patterns only
Common Sense	Strong, intuitive	Often weak or missing

Bias vs Fairness

Aspect	Bias	Fairness
Nature	A problem — unequal treatment	A goal — equal treatment
Source	Skewed or historically unjust data	Achieved through auditing, balanced data, monitoring

Prompt Sensitivity vs Prompt Engineering

Aspect	Prompt Sensitivity	Prompt Engineering
Nature	A problem — small wording changes shift output	A solution — deliberately crafting prompts for reliable results

Privacy vs Security

Aspect	Privacy	Security
Focus	Protecting personal/sensitive data	Protecting systems from attacks
Example Risk	Data leakage of user information	Prompt injection attack

CASE STUDIES

Real-Life Case Studies

ChatGPT Hallucination — Fake Legal Citations

What Happened: A lawyer used ChatGPT to help prepare a legal brief, which included citations to court cases that did not actually exist.
Could It Have Been Prevented? Yes — by independently verifying every citation against a real legal database before submission, and by using RAG-grounded tools designed for legal research.

Google Gemini — Inaccurate Image Generation Controversy

What Happened: An AI image generation feature produced historically inaccurate depictions in certain contexts, drawing public criticism.
Could It Have Been Prevented? Yes — through more careful testing of the model's behaviour across diverse prompts before public release.

AI-Generated Fake News

What Happened: AI text-generation tools have been used to mass-produce fake news articles that spread misinformation online.
Could It Have Been Prevented? Better platform-level detection of AI-generated content, content watermarking, and stronger misuse policies could reduce this risk.

Deepfake Videos

What Happened: Deepfake technology has been used to create fake videos of public figures saying things they never said.
Could It Have Been Prevented? Detection tools, platform moderation, and legal deterrents can help; media literacy education also reduces the impact.

Healthcare AI Errors

What Happened: Some AI diagnostic tools have shown reduced accuracy for underrepresented patient groups due to unbalanced training data.
Could It Have Been Prevented? Ensuring diverse, representative training datasets and requiring human doctor oversight before final decisions.

AI Hiring Bias

What Happened: A well-known case involved a hiring AI tool that learned to downgrade resumes containing certain gendered terms, reflecting biased historical hiring data.
Could It Have Been Prevented? Auditing training data for bias, testing hiring outcomes across demographic groups, and keeping human reviewers in the loop.

AI Recommendation Systems & Filter Bubbles

What Happened: Social media recommendation algorithms have been criticised for reinforcing narrow viewpoints or addictive engagement patterns.
Could It Have Been Prevented? Designing algorithms with diversity and wellbeing metrics, not just engagement/click-through rate, alongside more algorithmic transparency.

FINAL MIND MAP

Complete Mind Map — All Module 6 Concepts

This mind map connects every challenge covered in Module 6, branching out from the central theme of Generative AI's key limitations.

All 10 key challenges of Generative AI, connected to the central theme.

FINAL RECAP

One-Page Quick Revision Sheet

Core Challenges At a Glance

- **Prompt Sensitivity** — small wording changes shift AI output.
- **Knowledge Cutoff** — model's knowledge frozen at training date.
- **Non-Determinism** — same input can yield different outputs (sampling + temperature).
- **Hallucination** — confidently wrong, made-up content.
- **Lack of Common Sense** — no lived physical/social experience.
- **Bias** — unfair outcomes from skewed/historical data.
- **Privacy/Security/Misuse** — data exposure, attacks, and harmful use.

Fixes At a Glance

Problem	Key Fix(es)
Knowledge Cutoff / Hallucination	RAG, web search, fine-tuning, fact-checking
Non-determinism	Lower temperature / deterministic decoding
Bias	Data audits, fairness-aware algorithms, human oversight
Privacy/Security	Encryption, input validation, monitoring, red-teaming

Responsible AI Pillars

> Transparency • Accountability • Fairness • Privacy • Safety ("TAFPS")

Memory Tricks Recap

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Prompt Sensitivity: "Same meaning, different words → different answer"
Knowledge Cutoff: "Model's memory freezes on a date"
Hallucination: "Confident + Wrong"
Bias: "Biased data in → biased decisions out"
Responsible AI: "TAFPS" = Transparency, Accountability, Fairness, Privacy, Safety
```

Frequently Confused Concepts

- Hallucination (wrong content) vs Knowledge Cutoff (missing recent content)
- Bias (a problem) vs Fairness (the goal/solution)
- Privacy (protecting data) vs Security (protecting systems)
- AI Ethics (moral principles) vs AI Governance (enforcement policies)

Golden One-Liner

> "Generative AI is powerful but imperfect — it can misread prompts, forget recent events, answer inconsistently, hallucinate facts, miss common sense, reflect bias, and be misused — Responsible AI practices exist to manage every one of these risks."